

CLINICAL IMPROVEMENT IN SPINAL CORD INJURY TREATED WITH MESENCHYMAL STEM CELL: A CASE REPORT

CARLES SIAGIAN

INDONESIAN POLICE CENTRAL HOSPITAL

PCI-IOSS & ISASS-AP

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INTRODUCTION

Current Understanding, Challenge, and Potential
Future Option



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Introduction: Current Understanding

- Spinal Cord Injury (SCI) leads to changes in motor, sensory, or autonomic functions, with significant impacts on quality of life.
- Global Incidence: WHO estimates the global incidence of SCI at 40-80 cases per million annually, with 300,000 to 600,000 new cases worldwide.
- Causes: 90% of SCIs are caused by traumatic events, including traffic accidents, falls, and sports or work-related incidents.
- How is the treatment?



Introduction: Challenge and Potential Option

- Pharmacological Limitations: Medications like methylprednisolone and naloxone showed promise in animal models of SCI but failed in human trials.
- Stem Cell Potential: Stem cell therapies are emerging as a promising SCI treatment due to their potential to regenerate damaged neurological networks.
- Stem cell transplantation therapy, offering the potential to regenerate neurological networks, stands as a promising candidate for SCI treatment.

- Bracken, M.B., Sephard, M.J., Collins, et al. : A Randomized, Controlled Trial of Methylprednisolone or Naloxone in the Treatment of Acute Spinal Cord Injury. N Engl J Med. 322, 1405–1411 (1990)

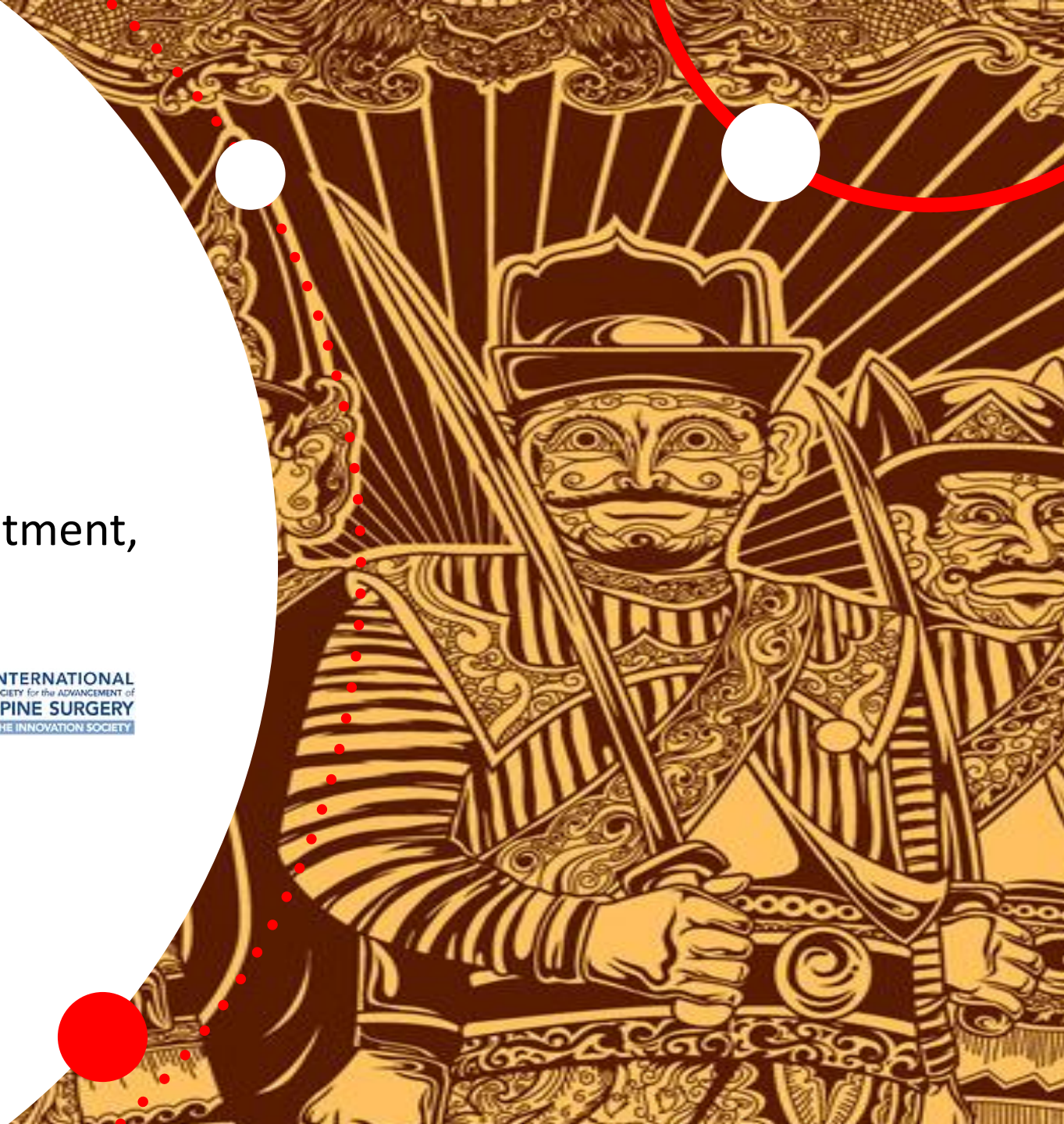
- Yamazaki, K., Kawabori, M., Seki, T., Houkin, K. : Clinical trials of stem cell treatment for spinal cord injury, (2020)

THE CASE

Patient's History, Findings, Diagnosis, Treatment,
and Follow Up



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The Case: Patient's History

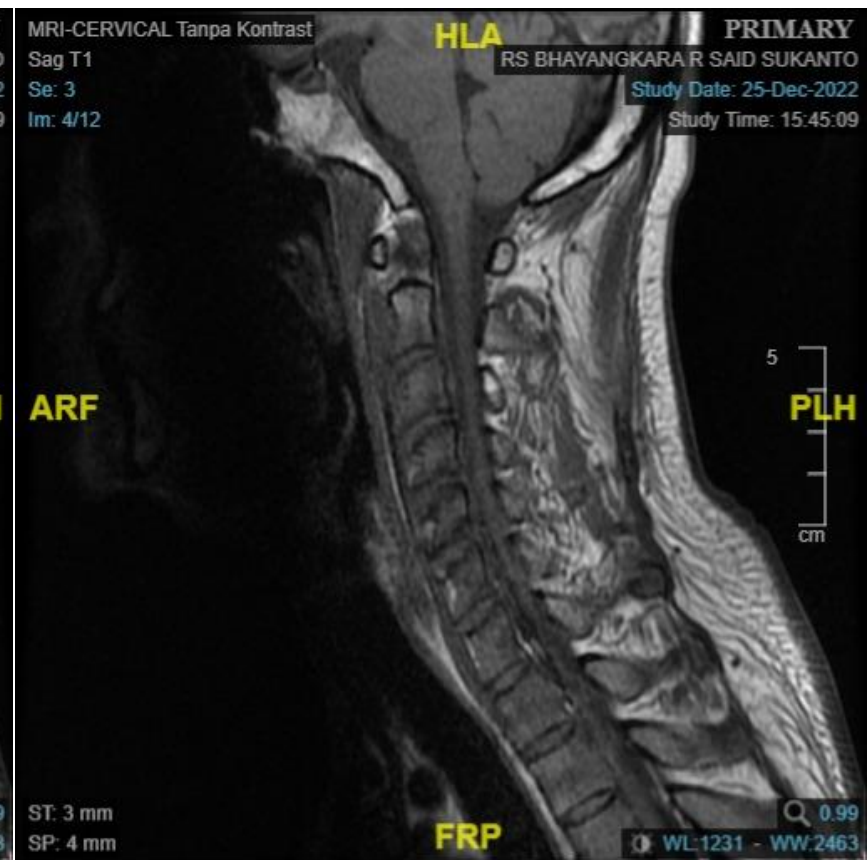
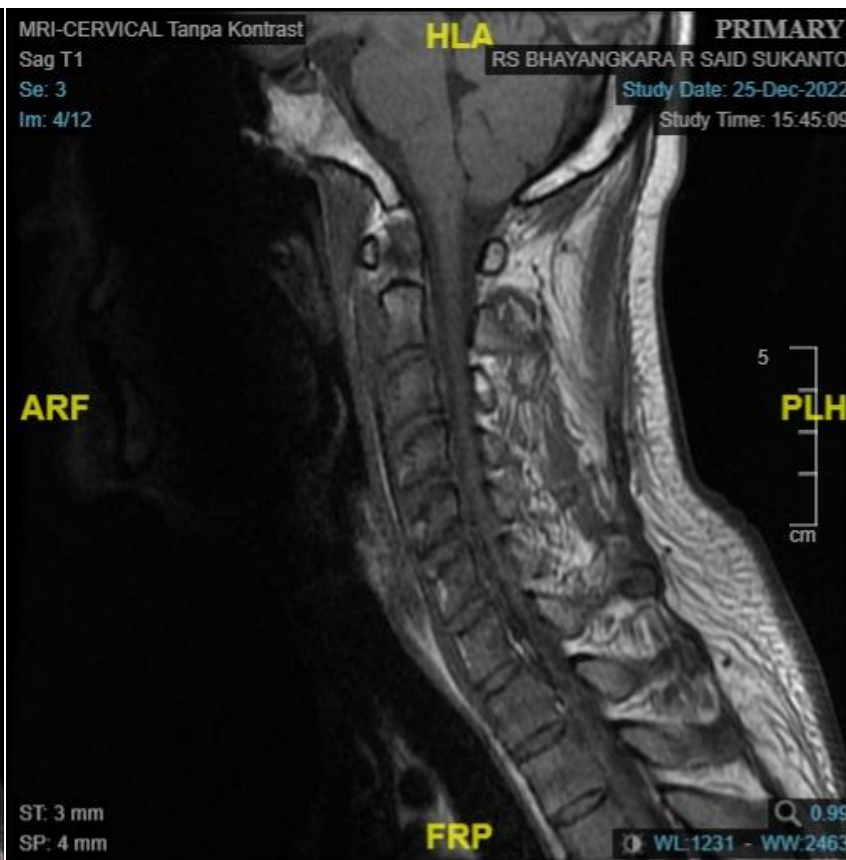
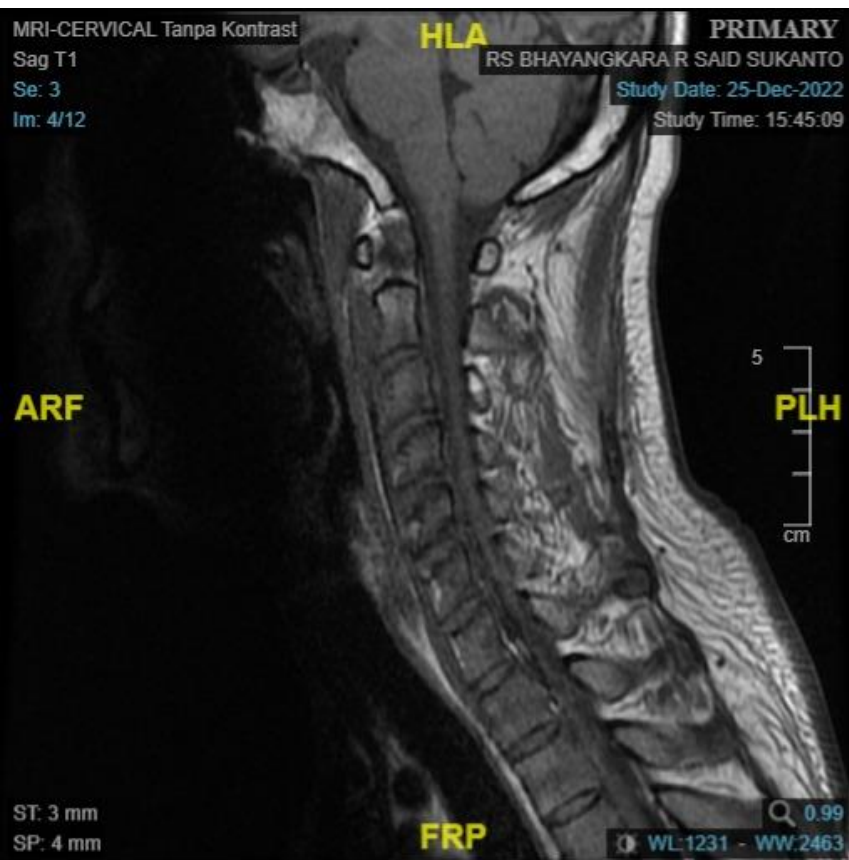
- A 45-year-old male police officer, with no significant medical history, presented to the emergency room (transferred from a satellite hospital) with a **week-long history of quadriplegia** following a traumatic head-first fall from a trail motorcycle, **impaired sensation** at C4, **slight motor function**, stable vital sign, **fully conscious**, stabilized with a neck collar and spine board.
- Suspected of cervical SCI, the patient remained tetraplegic with no motor improvement despite **a week of high-dose methylprednisolone** at a satellite hospital

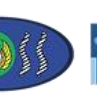


The Case: Radiology & Other Findings

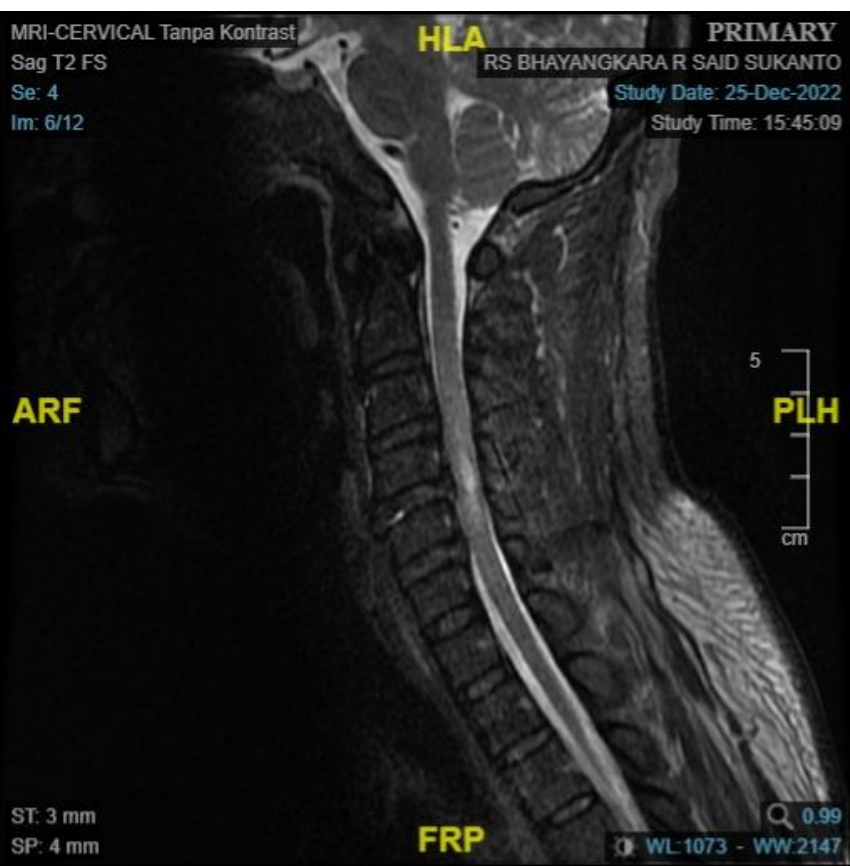
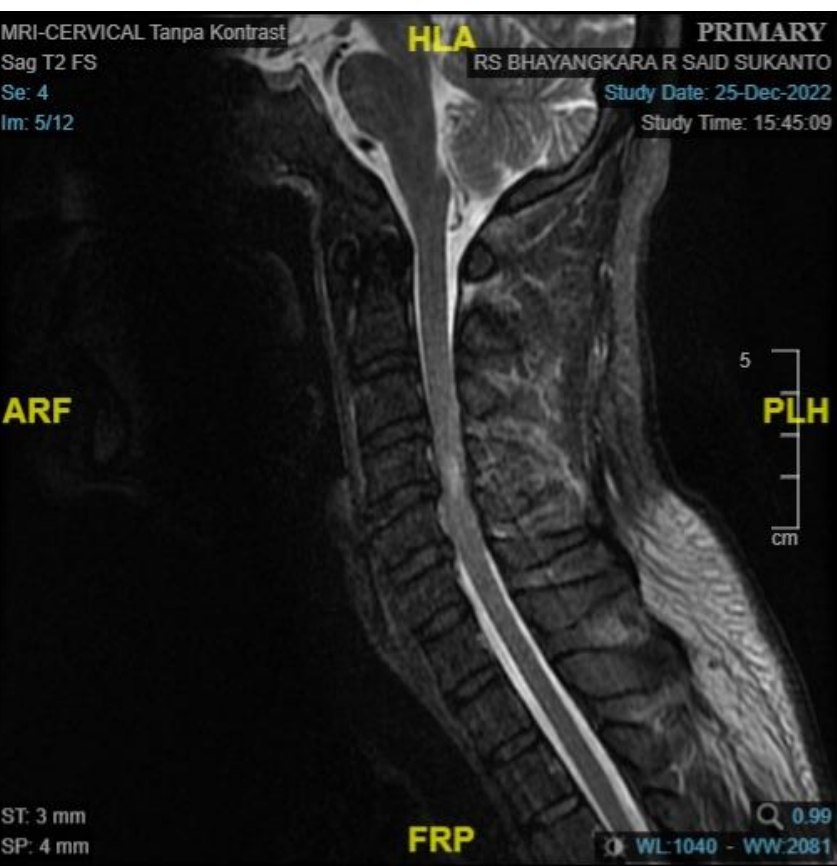
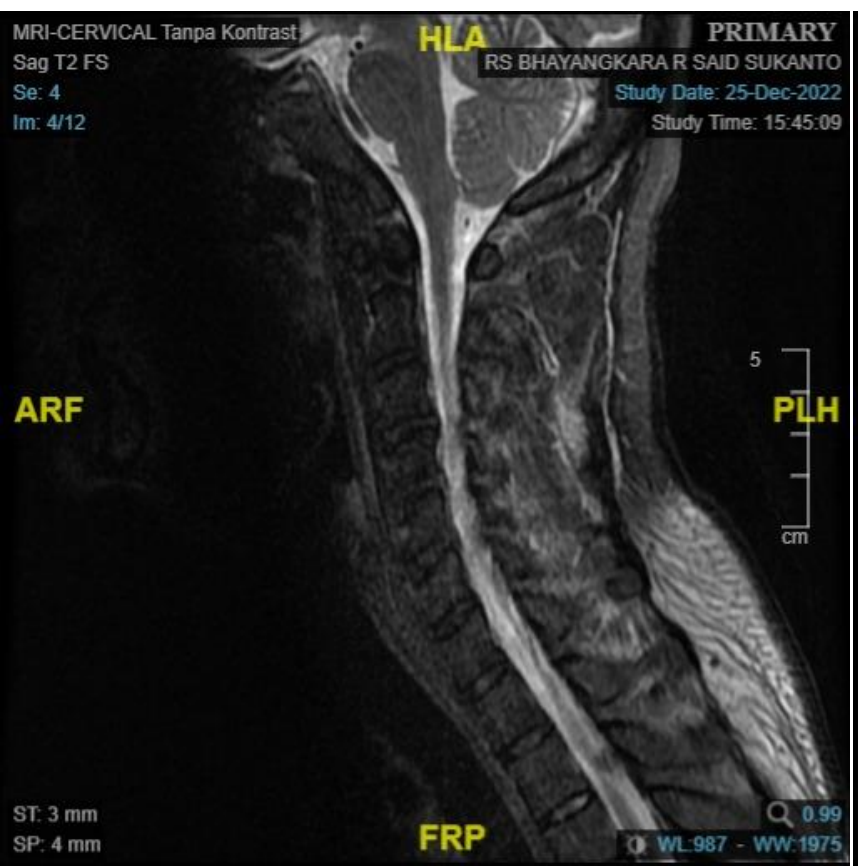
- A 3D head scan showed **no signs of bleeding or fractures** in the cranial base.
- MRI of the cervical spine identified **spinal canal stenosis from C4 through C6**.
- The EMG results showed reduced amplitudes in the bilateral median and peroneal motor nerves, as well as the left sural sensory nerve, with decreased conduction velocity was observed in the left ulnar motor nerve. No responses were elicited from the right sural sensory and left ulnar sensory nerves. These findings **suggest cervical sensory polyneuropathy** with radiculopathy.

The Case: Radiology Findings (MRI T1/Sagittal)





The Case: Radiology Findings (MRI T2/Sagittal)





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The Case: Diagnosis

The patient was diagnosed with:

Spinal Cord Injury ASIA Impairment Scale (AIS) C
with Neurological Level of Injury (NLI) at the Cervical C4 level.



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The Case: Dilemmas and Family Decision

In the initial weeks, the patient was informed about the diagnosis, patient's prognosis, and the **limited therapeutic options** in line with **refusal consent of surgery**, including the investigational nature of stem cell therapy, but chose to proceed the stem cell therapy option, fully supported by the patient's family.



The Case: Treatment Preparation

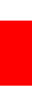
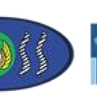
- **One month after the incident**, the patient planned to received mesenchymal stem cell therapy.
- The mesenchymal stem cells, **obtained from umbilical cord** sources.
- Contained of **50×10^6 viable cells** as requested one week prior, a total of 52.08×10^6 cells (**viability of 96%**).
- Prepared and packaged in sterile syringes containing a NaCl 0.9% as the medium solution.



The Case: Why Umbilical Cord?

- Umbilical cord-derived mesenchymal stem cells (UC-MSCs) have the capacity to differentiate into a uniform population of cells that not only express neural markers but also develop characteristics typical of neural phenotypes, and expressing CD73+ (rapid self-renewing), CD90+ (inflammatory responses, fibrotic progression, proliferation and differentiation), and CD105+ (angiogenesis, neovascularization).

Goodwin, H., Bicknese, A., Chien, S.-N., Bogucki, B., Oliver, D., Quinn, C., Wall, D.: Multilineage Differentiation Activity by Cells Isolated From Umbilical Cord Blood: Expression of Bone, Fat, and Neural Markers. *Biology of Blood and Marrow Transplantation*. 7, 581–588 (2001)



The Case: Stem Cells Profile

Immunophenotyping	Result	References
CD 73+	99.7%	$\geq 80\%$
CD 90+	98.2%	$\geq 80\%$
CD 105+	99.9%	$\geq 80\%$
Lin Neg	0.7%	$\leq 20\%$

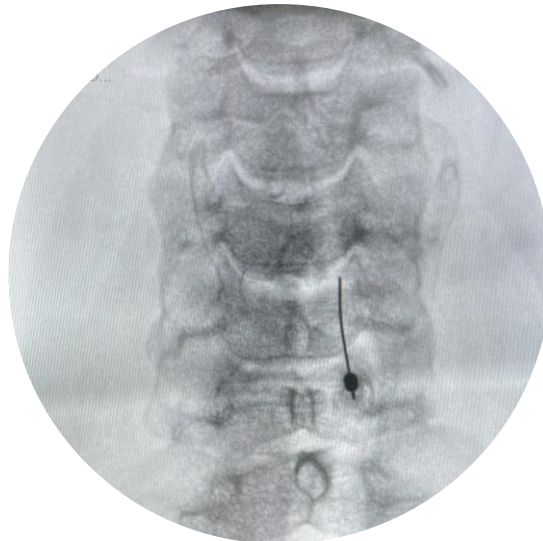
Microbiology Test	Result	References
Sterility	Negative	Negative
Endotoxin (LAL Test)	Negative	Negative (<0.03 EU/mL)
Mycoplasma (Chemiluminescence)	0.13	< 0.9 Negative $0.9 - 1.2$ Borderline > 1.2 Positive

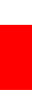
Screening Test	Result	Specification
HBsAg	Non-reactive	Non-reactive
Anti-HCV	Non-reactive	Non-reactive
Anti-HIV	Non-reactive	Non-reactive
TPHA	Non-reactive	Non-reactive
Anti-CMV IgM	Non-reactive	Non-reactive
Anti-CMV IgG	Non-reactive	Non-reactive
Anti-Toxoplasma IgM	Non-reactive	Non-reactive
Anti-Toxoplasma IgG	Non-reactive	Non-reactive



The Case: Intrathecal UC-MSCs Injection

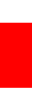
- The patient was sedated and prone positioned.
- C-Arm was used for precise localization.
- The mesenchymal stem cells were injected directly into the cervical intrathecal space between C4 and C5.





Motoric Improvement - Upper Muscle

Muscle Group	Motoric Level				
	Week 0	Week 1	Week 2	Week 3	Week 4
Upper Extremities					
Elbow Flexors - Right	0	2	3	4	4
Elbow Flexors - Left	1	3	3	4	5
Wrist Extensors - Right	1	2	3	4	5
Wrist Extensors - Left	0	3	4	5	5
Elbow Extensors - Right	1	3	4	5	5
Elbow Extensors - Left	0	2	3	5	5
Finger Flexors - Right	1	3	4	4	5
Finger Flexors - Left	1	3	4	5	5
Finger Abductors - Right	0	2	3	4	4
Finger Abductors - Left	1	3	3	4	5



Motoric Improvement - Lower Muscle

Muscle Group	Motoric Level				
	Week 0	Week 1	Week 2	Week 3	Week 4
Lower Extremities					
Hip Flexors - Right	2	3	4	5	5
Hip Flexors - Left	2	3	4	5	5
Knee Extensors - Right	1	2	3	4	4
Knee Extensors - Left	1	1	2	4	5
Ankle Dorsiflexors - Right	2	2	3	4	4
Ankle Dorsiflexors - Left	2	3	4	4	5
Long Toe Extensors - Right	1	3	4	5	5
Long Toe Extensors - Left	1	3	4	5	5
Ankle Plantar Flexor - Right	1	3	4	5	5
Ankle Plantar Flexor - Left	1	2	3	5	5



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Sensorics Improvement

Time	Neurological Level of Injury
Patient Admitted	Cervical C4
Post Therapy Day 1	Cervical C4
Post Therapy Day 2	Cervical C4-C5
Post Therapy Day 3	Cervical C7
Post Therapy Day 4	Thoracal T3
Post Therapy Day 5	Thoracal T6
Post Therapy Day 6	Thoracal T7-8
Week 1	Thoracal T12
Week 2	Lumbar L4
Week 3	Sacral S2
Week 4	Sacral S2



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The Case: Follow Up - Bladder and Defecating

- After stem cell therapy, bladder training was initiated, and the patient noticed urination sensation by the second week, eventually discontinuing the catheter.
- By the tenth day, residual urine dropped to less than 10%, allowing for independent urination.
- Three weeks post-therapy, the patient's defecation sensation returned, and bowel movements became regular.
- By the fourth week, the patient had improved balance and could defecate independently in toilet.



The Case: Follow Up





The Case: Rehabilitation Program

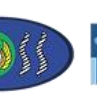
- Rehabilitation Protocol: mobilization, muscle strengthening, sensory re-education, bladder & bowel training, balance exercises, and static ergo-cycle cardiopulmonary endurance training using.
- By week 4, most muscle movements had improved, but the patient still felt their walking balance had not fully returned, the patient's hand grip strength did not fully recover, but elbow strength was sufficient to support body balance.
- The patient was provided with a single lofstrand crutch for the right arm on discharged.



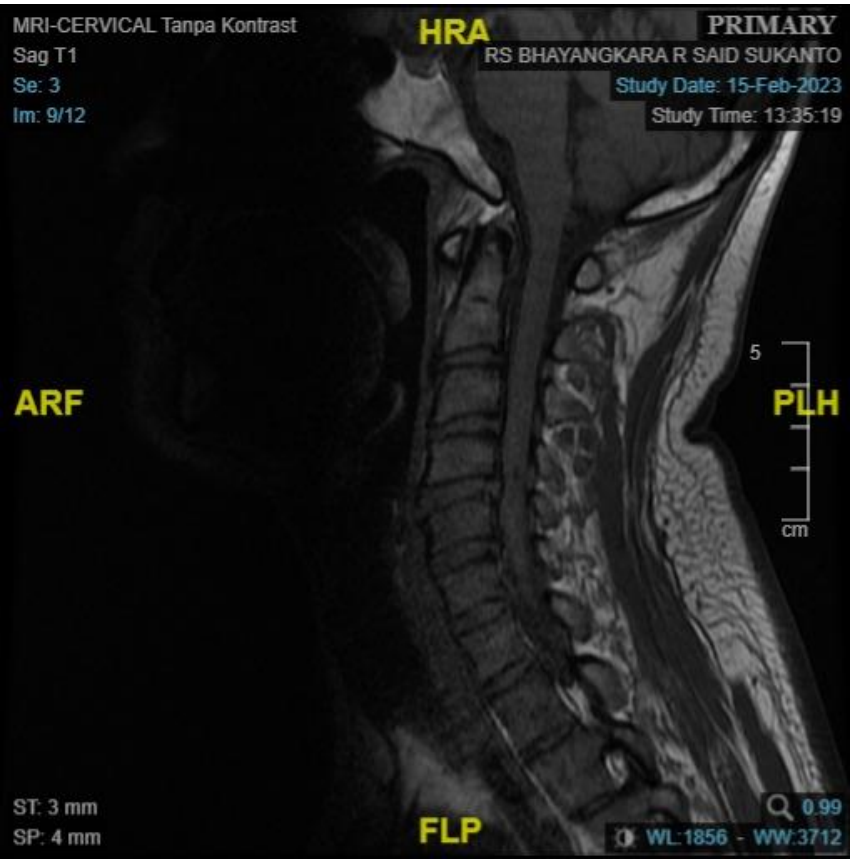
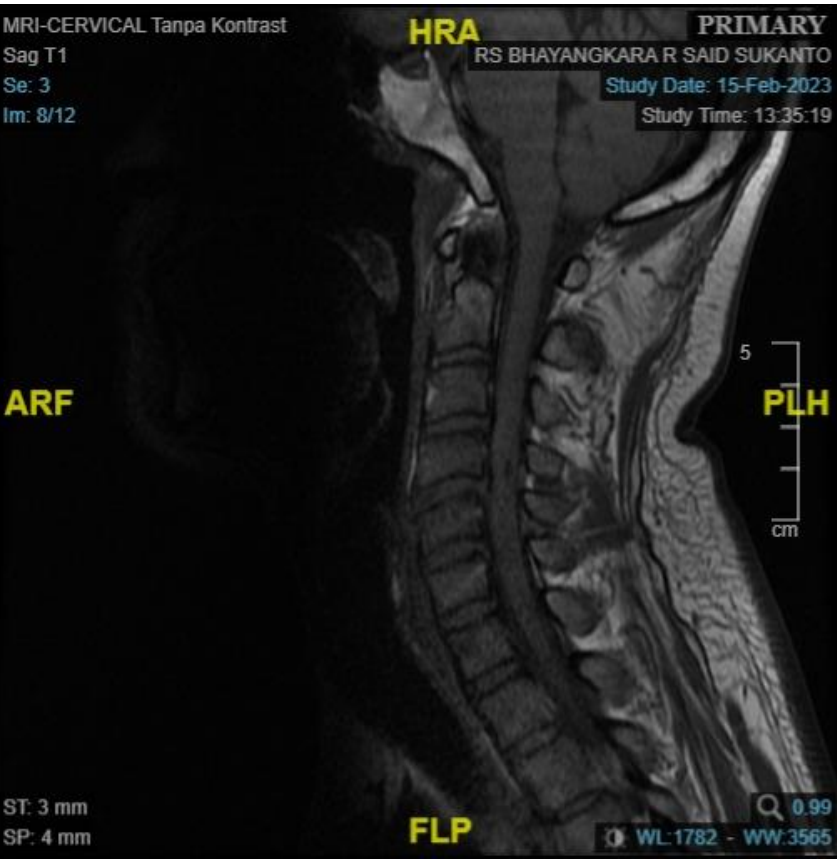
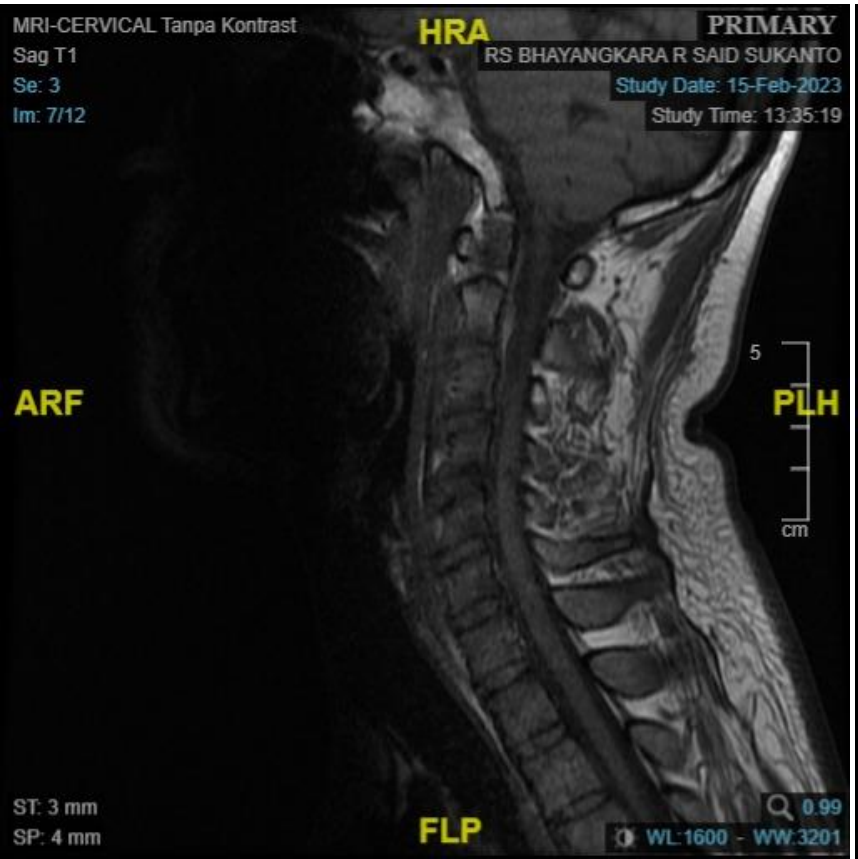
The Case: Discharge and Clinic Follow Up

- The patient was discharged one month after MSCs therapy.
- One month after discharge, the patient underwent cervical imaging examination
- Follow-up appointments were scheduled for the third, sixth, and ninth months at the clinic.
- The patient was able to perform activities independently without crutches, and without any movement restrictions after three and six months.



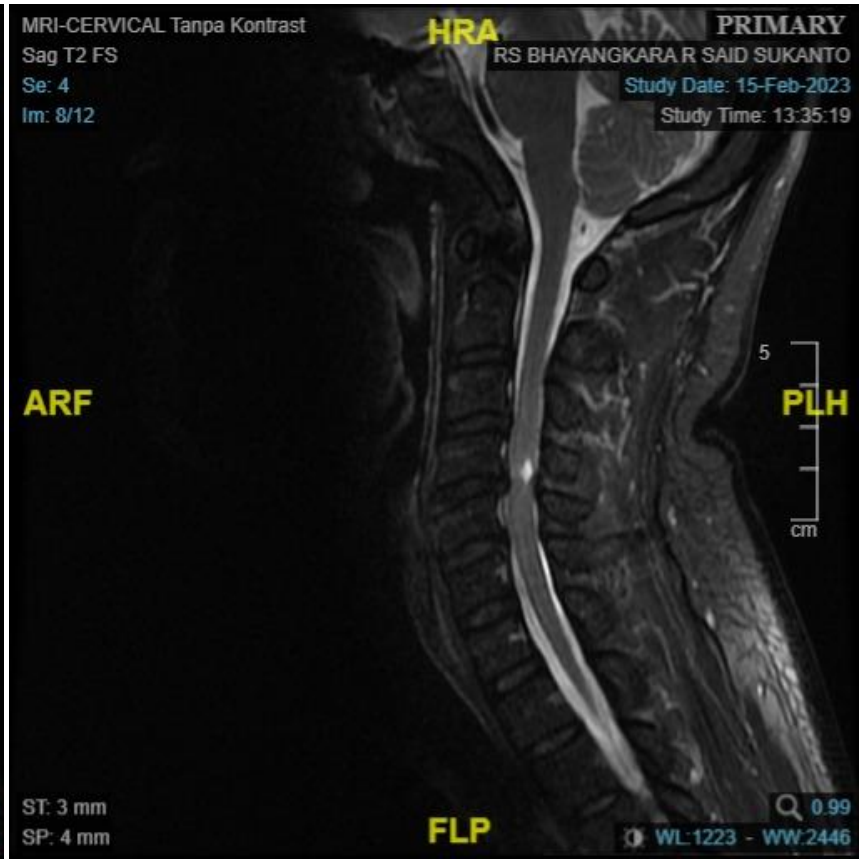
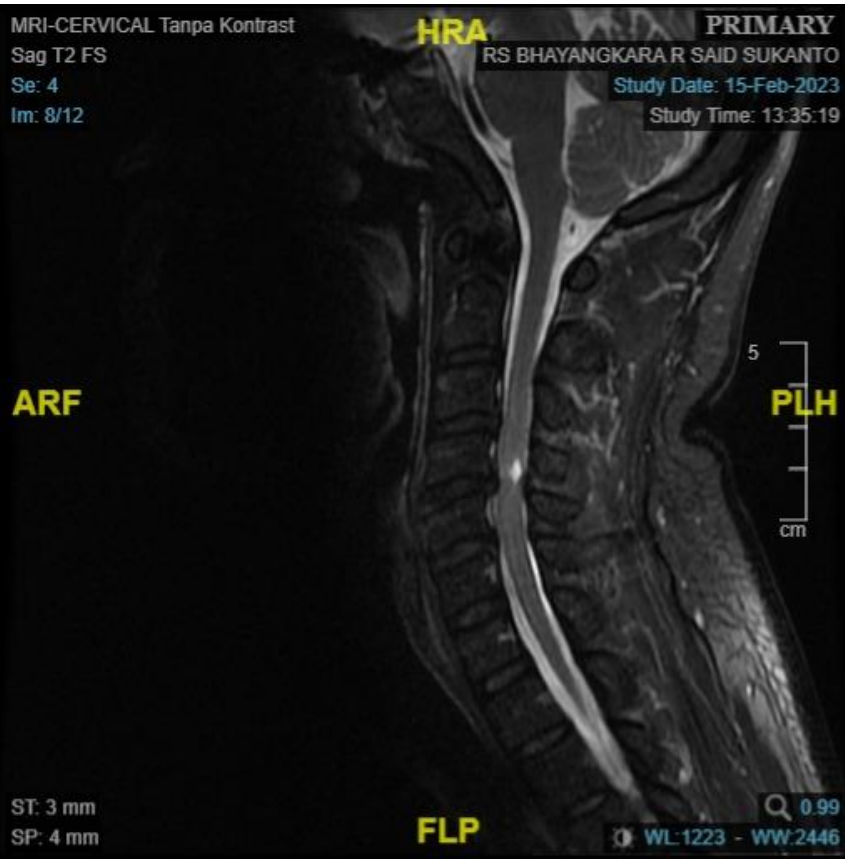
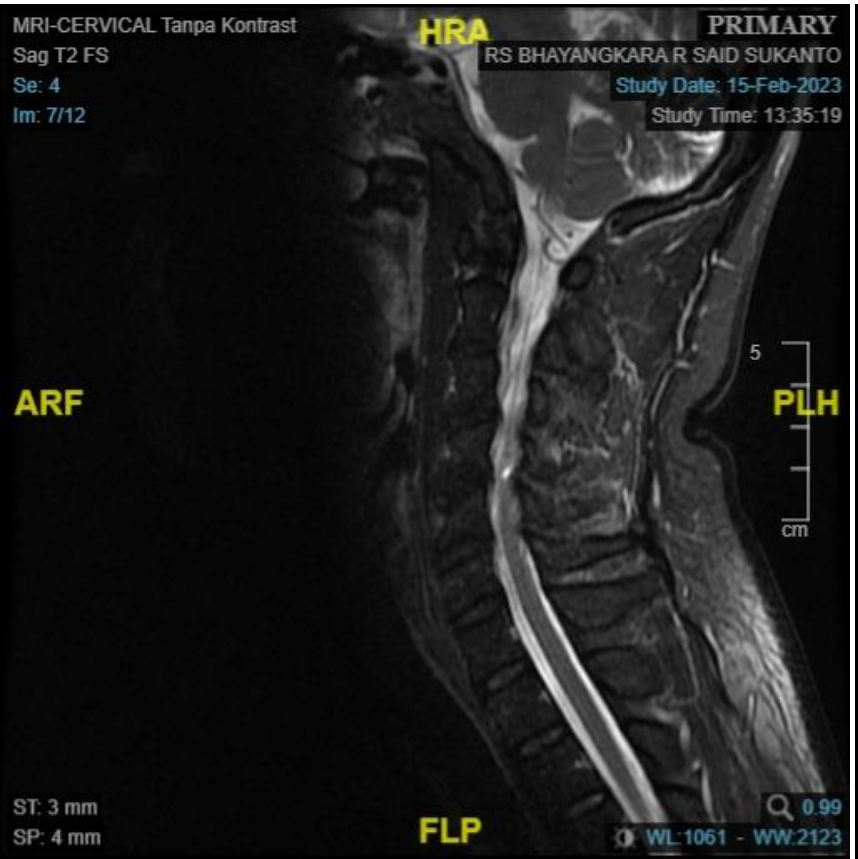


The Case: 1 Month Follow Up MRI T1/Sagittal





The Case: 1 Month Follow Up MRI T2/Sagittal





The Case: Lost to Follow Up?

- Although the patient could not attend the nine-month follow-up, they sent a video in the tenth month showing their return to work as an active police officer. The patient reported full mobility of all limbs, as they had prior to the incident, 12 months after stem cell therapy.



CONCLUSION



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Conclusion

- This case report highlights the promising role of mesenchymal stem cell therapy in addressing severe spinal cord injuries.
- The treatment not only restored motor functions, sensory perception, and bladder and bowel control but also enabled the patient to regain independence and return to work as before the injury.
- While additional research and larger clinical trials are necessary to confirm its efficacy and safety, our findings suggest that stem cell therapy could significantly enhance recovery and improve the quality of life for those with spinal cord injuries.



The Patient Perspective

- The patient reported that the stem cell therapy surpassed his expectations.
- He mentioned that, without the positive outcomes from the therapy, he would have been unable to live independently and might have had to leave his job as an active police officer.
- He expressed deep gratitude for his family's unwavering support, which included moral support, encouragement, religious guidance, and financial help. These elements were crucial and integral to his treatment process.

Thank You